QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

bdk @ f2528da88b10 · Bitcoin Dev Kit — public repository (passive review)

Organization	QSI
Report date	2026-10-28 19:24 UTC
Report version	1
Audit run	2b50b8d0-5f83-4b6a-8343-16d2e8f4851e · Document audit · 2026-10-28 19:24 UTC
Analyzer engine	qsi-engine 1.2.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`17cf808163d3356c7d53bcdb7c9e634e4a9e6400eb6e18601bfc81fadd83aba5`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

Contents

1. Executive summary
2. Scope and boundaries
3. Methodology
4. Assets examined and not examined
5. Architecture overview
6. Overall security posture
7. Findings by severity
8. Detailed findings
9. Evidence
10. Post-quantum assessment
11. Limitations and assumptions
12. Remediation plan
13. Priorities for 30, 60 and 90 days
14. Verification after remediation
15. Tools used
16. Sources and normative documents
17. Appendices

This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

1	0	0	1	0
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	0
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	1
Improvement recommendation	0
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-06106	Medium	Quantum-vulnerable ECDSA: transaction and message signing (1 file(s)) preliminary	35.6

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/bitcoindevkit/bdk @ f2528da88b1041bdddf547d677511aea19ad3573 — evidence links open the lines at this commit.

Supplied inputs

45 files, 592207 bytes in total, 0 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.2.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
dependencies	checked	7	crates/bitcoind_rpc/Cargo.toml: 4 dependencies parsed (0 pinned); crates/chain/Cargo.toml: 9 dependencies parsed (0 pinned); crates/core/Cargo.toml: 5 dependencies parsed (0 pinned)
source_code	checked	38	crates/bitcoind_rpc/src/bip158.rs: 197 lines scanned with 16 rule patterns and 122 registry patterns; crates/bitcoind_rpc/src/lib.rs: 627 lines scanned with 16 rule patterns and 122 registry patterns; 137 lines in #[cfg(test)] modules ignored; crates/chain/src/balance.rs: 105 lines scanned with 16 rule patterns and 122 registry patterns; 38 lines in #[cfg(test)] modules ignored
osv_advisories	not checked	1	no exactly pinned versions to query

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 45 (input files: C)

Not checked

- osv_advisories: no exactly pinned versions to query

Architecture overview

Static review of the source tree bdk at f2528da88b1041bdddf547d677511aea19ad3573. 45 files analysed; of 89 files, 30 are test, fuzzing, benchmark or example code and were left out, 14 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. No requests were sent to the owner's systems.

Overall security posture

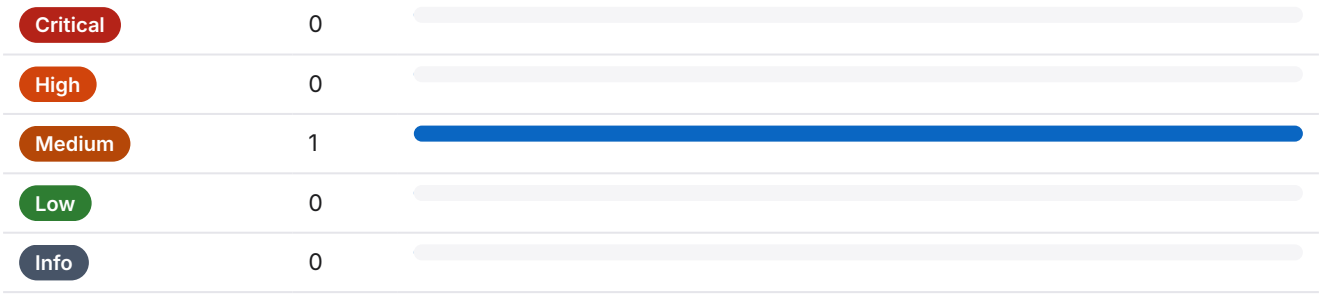
100
Indicative posture index (0–100)

100.0%
Quantum-vulnerable share of inventory

1
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-06106	Quantum-vulnerable ECDSA: transaction and message signing (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary

Detailed findings

QSI-2026-06106 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

2 reference(s) to ECDSA across 1 file(s), e.g. crates/chain/src/spk_iter.rs:2. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	crates/chain/src/spk_iter.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 19:24 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open crates/chain/src/spk_iter.rs:2 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · crates/chain/src/spk_iter.rs · line 2 · SHA-256 58953e8cd2768468… · File 5df96d8c7921b330…

```
1 | use crate::{
2 |     bitcoin::{secp256k1::Secp256k1, ScriptBuf},
3 |     miniscript::{Descriptor, DescriptorPublicKey},
```

file_excerpt · crates/chain/src/spk_iter.rs · line 40 · SHA-256 2100b069271cc491… · File 5df96d8c7921b330…

```
39 |     descriptor: D,
40 |     secp: Secp256k1<bitcoin::secp256k1::VerifyOnly>,
41 | }
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

Evidence

ID	Type	Data source	SHA-256
QSI-2026-06106	file_excerpt	crates/chain/src/spk_iter.rs (line 2)	58953e8cd27684683f75e8c61ebc1f8b6d9c286a2f5983b56e89a0ae255886b5
QSI-2026-06106	file_excerpt	crates/chain/src/spk_iter.rs (line 40)	2100b069271cc49191876e6a859414230e335185aeb5ec18d55ed509e55d5e14

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
transaction and message signing	ECDSA (1)	QSI-2026-06106	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
vulnerable (Shor)	1

Detailed findings: QSI-2026-06106

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDSA (1)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.
- osv_advisories — not checked: no exactly pinned versions to query

Remediation plan

QSI-2026-06106 · P3 · 90 days — short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06106 · P4 · 90 days — root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
None			

60 days

Priority	ID	Action	Type
None			

90 days

Priority	ID	Action	Type
P3	QSI-2026-06106	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-06106	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-06106	Re-run the inventory after refactoring.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.1.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.2.0
inventory_document	1.1.0
solidity	1.2.0
source_code	1.3.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.2.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

1. C. Chevignard, P.-A. Fouque, A. Schrottenloher — Reducing the Number of Qubits in Quantum Discrete Logarithms on Elliptic Curves (IACR ePrint 2026/280) — IACR ePrint. https://eprint.iacr.org/2026/280 (revision of a EUROCRYPT 2026 paper: 3.12n + o(n) qubits; for n = 256, 1193 qubits and 22 runs of 2^38.98 Toffoli gates)
2. Common Vulnerability Scoring System v3.1: Specification Document — FIRST. https://www.first.org/cvss/v3.1/specification-document (final)
3. D. Litinski — How to compute a 256-bit elliptic curve private key with only 50 million Toffoli gates (arXiv:2306.08585) — arXiv. https://arxiv.org/abs/2306.08585 (preprint (2023-06-14): 6000 modules × 1152 physical qubits, one key per 10 minutes under its architecture assumptions)
4. FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM) — NIST. https://csrc.nist.gov/pubs/fips/203/final (final (August 2024))
5. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
6. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
7. J. Proos, C. Zalka — Shor's discrete logarithm quantum algorithm for elliptic curves (Quantum Inf. Comput. 3(4), 2003; arXiv:quant-ph/0301141) — Quantum Information & Computation. https://doi.org/10.26421/QIC3.4-3 (peer-reviewed article)
8. M. Cain et al. — Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits (arXiv:2603.28627) — arXiv. https://arxiv.org/abs/2603.28627 (preprint (2026-03-30): P-256 in a few days with 26,000 physical neutral-atom qubits; RSA-2048 one to two orders of magnitude longer)
9. M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018) — IEEE. https://doi.org/10.1109/MSP.2018.3761723 (peer-reviewed article; source of the x + y > z migration inequality)
10. M. Roetteler, M. Naehrig, K. Svore, K. Lauter — Quantum resource estimates for computing elliptic curve discrete logarithms (ASIACRYPT 2017) — IACR / arXiv. https://arxiv.org/abs/1706.06752 (peer-reviewed article: at most 9n + 2⌈log2 n⌉ + 10 qubits and 448 n^3 log2 n + 4090 n^3 Toffoli gates)
11. NIST IR 8547 — Transition to Post-Quantum Cryptography Standards — NIST. https://csrc.nist.gov/pubs/ir/8547/ipd (initial public draft (November 2024); proposes deprecating quantum-vulnerable public-key algorithms after 2030 and disallowing them after 2035 — dates may change in the final version)
12. P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997) — SIAM. https://doi.org/10.1137/S0097539795293172 (peer-reviewed article)
13. R. Babbush et al. — Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations (arXiv:2603.28846) — arXiv. https://arxiv.org/abs/2603.28846 (preprint (2026-03-30, v2 2026-04-15): 256-bit ECDLP with < 1200 logical qubits and < 90 M Toffoli gates (or < 1450 and < 70 M); < 500,000 physical qubits at 10^-3 error, minutes)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Location	Quantum status	Classical status
ECDSA	secp256k1	crates/chain/src/spk_iter.rs:2	vulnerable (Shor)	acceptable

B. JSON

qsi.report/1 · SHA-256 17cf808163d3356c7d53bcdb7c9e634e4a9e6400eb6e18601bfc81fadd83aba5

C. Input files

File	Size	SHA-256	Redactions
crates/bitcoind_rpc/Cargo.toml	1205	83b91c7a3e4ca359834c618240fa8f3a0a5e758f6732d666046c41a11bf49d1e	0
crates/bitcoind_rpc/src/bip158.rs	6761	b34cdc094a2395ce3336c503df9bcc0ee29f344f741ab25a3392d80020c1f06e	0
crates/bitcoind_rpc/src/lib.rs	24629	baf69cb786789de4b5f1b9357cd496200656552d93bf3ae8958180b289c7d41e	0
crates/chain/Cargo.toml	1488	0a170fa2a682fa7b5dd32a1a18d3c923136b75e44bf0bbc6497a6a565d8b38b3	0
crates/chain/src/balance.rs	3239	c4722689b26a97489acec069808665f2d99f21756d149dd98526a622fb596188	0
crates/chain/src/canonical.rs	28704	6d3081611bb7099213687c0ec0225ace91dd7662cdd20161b30cd37d16c34022	0
crates/chain/src/canonical_task.rs	19546	a774ade1336130608faff6ce1ecca7b07b9665c1a4c00eeb6a29fa0691d525ef	0
crates/chain/src/canonical_view_task.rs	8775	f60cac7e957c47be21c1e66c8a75230b8f1bea954d6784716a24ca05c9afcb32	0
crates/chain/src/chain_data.rs	13411	770bdca6ee7e8706a14a4ab08f93fb094d2c7ba5149452d23f75d71f3adf64c7	0
crates/chain/src/descriptor_ext.rs	1395	c046997e44f8ebdf762a39d259763091cd70ec1897e22cc7e6ac253705bdf837	0
crates/chain/src/example_utils.rs	2544	8f76a44443ebc40cceff4833b82898a1eda21784774582649e4a3edd2d80c710	0
crates/chain/src/indexed_tx_graph.rs	20256	9d90f8c1ce2109d50a351f4c8c9f7e360bde210957cfc95f825ffedc509226ad	0
crates/chain/src/indexer/keychain_txout.rs	55225	13f5011338a01c5776262b07538efd04338f13c9ad33349a06c925b5d38a8539	0
crates/chain/src/indexer/spk_txout.rs	19747	3f60ae8ee0ec261ba9a93407ef2d9e232787526bf4fad22daf6877e9ca1f1820	0

crates/chain/src/indexer.rs	2332	2745f08c61b5c64bad7768ae5a5f5cd94bbf608120f62df8ff0fead96b227653	0
crates/chain/src/lib.rs	2931	90a0f842b2b47069220959344f84c0aa4c8ecfc1344bff8572dd2e307e176d9e	0
crates/chain/src/local_chain.rs	35484	81ff161a87ed2621837edac91ef5f4b44e01d8ad2cdb2fcd2a7e71c64b69fc1f	0
crates/chain/src/rusqlite_impl.rs	31001	f0d1ee07c37b7463e0d6d51509518fb765da6a043a0f80ef5253d8dd7e39a8da	0
crates/chain/src/spk_iter.rs	11496	5df96d8c7921b3306d986d117a4348e9afe98be7c27c94b1ef8fe1ab13c2f7b9	0
crates/chain/src/tx_data_traits.rs	3937	3955c21e9df3f266e015156de68802b855c4eff8b4248dcb2beba6ccf3ca9748	0
crates/chain/src/tx_graph.rs	54561	6e99486950e6e752f1451b68fef0df8f2b365c4c816c33aac9b263e6a75bf317	0
crates/core/Cargo.toml	967	b7d7c5845ff0fe6681193d2de326a75c076e01201a1a7ee345dda16980f1cb28	0
crates/core/src/block_id.rs	1431	9b95ef3640317298b5a38d6fa7b89fdac18edc53aac55292b7a85f0f4f850619	0
crates/core/src/chain_query.rs	1659	369fb027368651a610f09a66bce4adaf2fc7774eeff122ab6a368a3b0471ac6b	0
crates/core/src/checkpoint.rs	26783	fbb0c2a3f0d13941adc50f6d8ba9f84cd703d1505f51a098dd4fa011acc01f4f	0
crates/core/src/checkpoint_entry.rs	5428	6864fd17b358f33675409d53e65ff00d83cd929d507c600bf1dbfb1f80c7cb76	0
crates/core/src/lib.rs	2986	70bd648e7f01758a128bd04cce0251d5194bd1e84b9ae4a9ac90a0e6d9f5e327	0
crates/core/src/merge.rs	2522	a7d01b0607e0b525e9841d59a2d011bd2e8d2edb807a888634db7400922b5c49	0
crates/core/src/spk_client.rs	26500	b36f7eac9ea9faef7ca22e2604d857d218ed1b1aa47a2657edb659e5a501b91d	0
crates/core/src/tx_update.rs	4343	4fa8a8f174889198258acc3293f726142d6c1882c41468be839c4db8077e2af7	0
crates/electrum/Cargo.toml	984	71d66c3d4f5fad2b279c8ce0d48ada975679f93668f9425670e14d1fe19b0200	0
crates/electrum/src/bdk_electrum_client.rs	36790	e390a2b8437c83a9d14daeb3f923b19797efea2fb50e6a11fb8f79f58b12f53b	0
crates/electrum/src/lib.rs	1100	bbbeb08934d653c218be80344abe9f46b62085154b47ca29fe0b704f1370d3c0	0
crates/esplora/Cargo.toml	1748	1daccf1d0415e843e617673ccedabfb49813555855b01c80f8cf9b3853a2acda	0
crates/esplora/src/async_ext.rs	28886	c2fb578219b5fc719a943f6529f68b25857c9191c866752f283b005c31698aed	0

crates/esplora/src/blocking_ext.rs	35556	dd5711d633b794f71c5b5ce84767028799dd1f8d9f61cc01d1b995a0f15a27e0	0
crates/esplora/src/lib.rs	2777	69cd2fb696e1b267c0891e57eadac8ada2f1a27fe96305e6889930569f9580ff	0
crates/file_store/Cargo.toml	761	afcf61e36799750107cce36d0333badc52cbaaf3803d7eca9ec35e164de765d4	0
crates/file_store/src/entry_iter.rs	2635	00cb247c7ca63d3ae44317c2abbaffdd5b36a49bfbbf99ff5fca9c79234a299b	0
crates/file_store/src/lib.rs	1714	1010d865adab4b312801b24b93f4f851d6da991ba31e449907aba70aad735550	0
crates/file_store/src/store.rs	23655	9390058ea9f9ab92fe1ca6732e3dbf06fd99822533b48b5a91d5a98ae46fcfce	0
crates/testenv/Cargo.toml	1067	1595b83bbd626078abc111bd7ac9233df0406f944473aa8d0e1a956de0dc4845	0
crates/testenv/src/lib.rs	21293	16bd1975ff6ead0e1f8bb86c65f33bec630835b2f21f48ff6c61e4dde0ea97aa	0
crates/testenv/src/persist_test_utils.rs	6370	fcfef6b7701ba51df6048b7ddd9ace6402ff9fe29cd7b8e0eacf16b77d8e554f	0
crates/testenv/src/utils.rs	5585	50b9ebfe246c3d0be4e277d2acd16773aabe06d1bfff2ee1e87f38efe851b894	0